

## 🐼 M003 Extension — Build the Panda (Intermediate)

**Topic:** 3D Panda (x, y, z) · **Course:** 3D Coordinates · **Level:** Intermediate (Extension, after Lesson 6) · **Time:** about 40 minutes

Build a **standing panda** in 3D — white body, black ears, black eye patches, black legs. Every block needs **three** numbers (**x, y, z**): x **across**, y **up**, z **deeper** (forward).

- Stand on your **home spot** (red block) every run. Move your feet, move your panda.
- 🎨 White parts = **white concrete**. Black parts = **black wool**.

### 1 · Predict 🧠

Read each set. Write a sentence for each.

```
place black block at (2, 3, 1)
place black block at (2, 4, 1)
place black block at (2, 5, 1)
```

**Only y changes. What happens to this leg?**

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```
place white block at (2, 8, 1)
place white block at (2, 8, 2)
place white block at (2, 8, 3)
```

**Only z changes. What happens to the line?**

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```
place white block at (2, 7, 1)
place black block at (2, 7, 1)
```

**Same (x, y, z). What do you see, and why?**

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## 2 · Spot the Bug 🐛

Fix each one. Then say why it was wrong.

**Bug A** — This ear should sit one step **deeper**, at (2, 9, 2). It has only two numbers, so it lands flat on the front.

```
place black block at (2, 9)
```

**Write the fixed code:**

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**Why was it wrong? Why does your fix work? (1–2 sentences)**

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**Bug B** — This eye should be **up high on the front**: (5, 7, 1). The last two numbers are swapped.

```
place black block at (5, 1, 7)
```

**Write the fixed code:**

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**Why was it wrong? Why does your fix work? (1–2 sentences)**

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### 3 · Show Your Work 📷 🎥

Look at the 🐼 panda in the world. Walk around it. Then build **one full panda** on your home spot — head, two ears, two eyes, two legs.

**Front face** (keep  $z = 1$ ): fill a white head from (2, 5, 1) to (5, 8, 1). Add two black ears on top — (2, 9, 1) and (5, 9, 1). Add two black eyes — (2, 7, 1) and (5, 7, 1). Add two black legs at the bottom — fill (2, 3, 1) to (2, 4, 1) and (5, 3, 1) to (5, 4, 1).

**Add depth:** build the white head again at  $z = 2$ , then  $z = 3$ , until your panda is as deep as the real one. Build the ears and legs deeper too.

**Face on front only:** at  $z = 2$  and  $z = 3$ , place plain white where the eyes were.

💡 Tip: the eyes only show on the **front**. The sides and back are plain white.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera at the panda. Name the parts.

Fill the blanks:

Today I built \_\_\_\_.

I built it using this code: \_\_\_\_.

In this code I used \_\_\_\_.

The hardest part was \_\_\_\_.

That part was hard because \_\_\_\_.

The most fun part was \_\_\_\_.

Something new I learned was \_\_\_\_.

**Write your lines here, then say them in your video.**

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**Submit** 

Send this worksheet + your **one** video to teacher on KakaoTalk.